

DEBRAJ DAS

Kolkata, West Bengal

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Education

Jadavpur University **Kolkata, India** *Bachelor of Engineering in Construction Engineering (CGPA: 7.8)*
2024 – 2028

WWA Cossipore English School **Kolkata, India** *Class 12 (ISC) – Percentage: 94%*
2024

St. Xavier's Institution **Kolkata, India** *Class 10 (ICSE) – Percentage: 93.5%*
2022

Technical Skills

Languages & Databases: Python, C++, JavaScript, SQL, PostgreSQL, HTML, CSS

Frameworks & Libraries: FastAPI, React, Node.js, Express.js, SQLAlchemy, Tailwind CSS

AI/ML & Architecture: XGBoost, Isolation Forest, SHAP, LLMs (Groq, Llama 3), REST APIs

Tools & Fundamentals: Docker, Git, VS Code, DSA, DBMS, OOP

Projects

Full-Stack Algorithm Visualizer (Vanilla Javascript, NodeJS, ExpressJS) | Github | Website

- Developed an interactive platform visualizing 12 algorithms across sorting, searching, and pathfinding, including BFS, DFS, Dijkstra and A*.
- Built REST APIs with Node.js and Express.js to execute algorithms and generate step-by-step execution traces in JSON.
- Created a Vanilla JavaScript frontend with animated visualizations, playback controls, statistics, and synchronized pseudocode highlighting.
- Implemented run-history persistence, input validation, error handling, CORS, and execution-time tracking.

AI Interview Prep SaaS (FastAPI, Groq's Llama 3.3, Docker) | Github | Website

- Built and deployed a full-stack platform integrating Groq's Llama 3.3 for AI-generated interview questions, automated LLM-as-judge answer scoring, and synthesized performance reviews
- Designed a multi-tenant REST API (FastAPI + SQLAlchemy) with JWT authentication, enforcing per-user data isolation across a 5-table relational schema.
- Built a live, multi-language code execution feature (Python/JS/C++/Java/C) for technical questions, integrating a third-party compiler API with graceful error handling.
- Developed a React frontend with real-time scoring visualization and a full-flow answer/review UX; containerized the backend with Docker and deployed independently via Render and Vercel

BankGuard (Python, Machine Learning (XGBoost / SHAP), React, FastAPI) | Github | Website

- Built a full-stack digital banking platform (React, FastAPI, PostgreSQL) with role-based auth, atomic transaction handling, and a live UPI payment integration via Razorpay.
- Designed and deployed two production ML models — an XGBoost + Isolation Forest fraud detection system (ROC-AUC 0.82) and a SHAP-explainable credit risk model — both driving real-time approval decisions, not just static predictions
- Engineered an admin console with fraud investigation workflows, audit logging, and a unified risk-scoring engine combining fraud, credit, and account-level signals.
- Deployed the application live (Render + Vercel) using Docker Compose, with automated database initialization and CI-ready health checks.

Achievements

- Achieved an All India Rank of 2368 in West Bengal Joint Entrance Examination (WBJEE), among 120,000 candidates.
- Reached the highest rating of 1350(1-star) on CodeChef.
- Reached the specialist on Codeforces (1421 rated).
- Solved over 500 problems on diverse coding platforms

Profile Links

- [Codeforces](#) | [CodeChef](#) | [Github](#)